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United States Patent	12419167
Kind Code	B2
Date of Patent	September 16, 2025
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Display device

Abstract

A display device includes: a substrate having a display area and a non-display area adjacent to the display area; a first conductive layer disposed on the substrate and including first conductive line; a first insulating layer disposed on the first conductive layer; a second conductive layer disposed on the first insulating layer and including a second conductive line, wherein, corresponding to the display area, the second conductive line and the first conductive line cross from a top view; a second insulating layer disposed on the second conductive layer; and a third conductive layer disposed on the second insulating layer and including a third conductive line; wherein, corresponding to the non-display area, a portion of a projection of the third conductive line on the substrate is overlapped with a portion of a projection of the first conductive layer on the substrate.

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Appl. No.: 18/598401

Filed: March 07, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240215360 A1	Jun. 27, 2024

Foreign Application Priority Data

CN	201810158603.2	Feb. 22, 2018
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Related U.S. Application Data

continuation parent-doc US 18314354 20230509 US 11957018 child-doc US 18598401
continuation parent-doc US 17399529 20210811 ABANDONED child-doc US 18314354
continuation parent-doc US 16259352 20190128 US 11121201 20210914 child-doc US 17399529

Publication Classification

Int. Cl.: **H10K59/131** (20230101); **G09F9/33** (20060101); **H05K1/02** (20060101); **H05K1/11** (20060101); **H05K3/06** (20060101); **H10D86/40** (20250101); **H10D86/60** (20250101)

U.S. Cl.:

CPC **H10K59/131** (20230201); **G09F9/33** (20130101); **H05K1/025** (20130101); **H05K1/115** (20130101); **H05K3/06** (20130101); **H10D86/441** (20250101); **H10D86/60** (20250101); **H10K59/1315** (20230201); G09G2300/0426 (20130101); H05K2201/09227 (20130101); H05K2201/10128 (20130101)

Field of Classification Search

CPC: H05K (1/025); H05K (1/115); H05K (3/06); H05K (2201/09227); H05K (2201/10128); H01L (27/124); H01L (27/3276); H01L (27/3279); G09G (2300/0426); G09F (9/33); H10K (59/131); H10K (59/1315); H10D (86/411); H10D (86/60)

USPC: 345/55; 345/204; 174/251

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefits of the Chinese Patent Application Serial Number 201810158603.2, filed on Feb. 22, 2018, the subject matter of which is incorporated herein by reference. (2) This application is a continuation (CA) of U.S. Patent application for “DISPLAY DEVICE”, U.S. application Ser. No. 18/314,354 filed on May 9, 2023; U.S. application Ser. No. 18/314,354 is a continuation (CA) of U.S. Patent application for “DISPLAY DEVICE”, U.S. application Ser. No. 17/399,529 filed on Aug. 11, 2021; U.S. application Ser. No. 17/399,529 is a continuation (CA) of U.S. Patent application for “DISPLAY DEVICE”, U.S. application Ser. No. 16/259,352 filed Jan. 28, 2019, and the subject matter of which is incorporated herein by reference.

BACKGROUND

1. Field

(1) The present disclosure relates to display devices and, more particularly, to a display device which, in the case of the presence of a non-display area on a display panel or the presence of an opening on a substrate, has thin bezels.

2. Description of Related Art

(2) Owing to technological advancement, electronic products are increasingly lightweight, thin or compact in order to meet consumer needs. Hence, display devices nowadays are no longer restricted to cathode ray tubes (CRT) but develop into thin display units. The thin display units can be applied to cellular phones, cameras, camcorders, laptops, mobile navigation devices, vehicle dashboards, electronic watches, and TV sets. Related manufacturers not only endeavor to develop thinner display devices but also attach great importance to thin bezel design.

(3) However, wirings and components will surround a non-display area or an opening in the event of the presence of the non-display area on a display panel or the presence of the opening on a

substrate. Furthermore, owing to process-related limitation, the non-display area is unfit for thin bezel design and thus fails to meet consumer needs.

(4) Therefore, it is imperative to provide a display device that overcomes wiring-related limitation and thus has thin bezels even when a display panel comprises a non-display area or even when a substrate comprises an opening.

SUMMARY

(5) The present disclosure provides a display device comprising: a substrate comprising a display area and a non-display area adjacent to the display area, a first conductive layer disposed on the substrate and comprising first conductive line; a first insulating layer disposed on the first conductive layer; a second conductive layer disposed on the first insulating layer and comprising a second conductive line, wherein, corresponding to the display area, the second conductive line and the first conductive line cross from a top view; a second insulating layer disposed on the second conductive layer; and a third conductive layer disposed on the second insulating layer and comprising a third conductive line; wherein, corresponding to the non-display area, a portion of a projection of the third conductive line on the substrate is overlapped with a portion of a projection of the first conductive layer on the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a top view of a portion of a display device according to an embodiment of the present disclosure;
- (2) FIG. 1B is a cross-sectional view of the display device taken along line M-M' of FIG. 1A;
- (3) FIG. 1C is a top view of a portion of the display device according to an embodiment of the present disclosure;
- (4) FIG. 2A is a top view of a portion of the display device according to an embodiment of the present disclosure;
- (5) FIG. 2B is a cross-sectional view of the display device taken along line N-N' of FIG. 2A;
- (6) FIG. 3 is a cross-sectional view of a portion of the display device according to an embodiment of the present disclosure; and
- (7) FIG. 4 is a cross-sectional view of a portion of the display device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) The implementation of the present disclosure is hereunder illustrated by specific embodiments. Persons skilled in the art can easily understand the other advantages and effects of the present disclosure by referring to the disclosure contained herein. The present disclosure can also be implemented or applied by the other different specific embodiments. Depending on different viewpoints and applications, various modifications and changes can be made to the details disclosed herein without departing from the spirit of the present disclosure.

(9) Ordinal numbers, such as 'first', 'second' and 'third', used hereunder describe claimed components but do not state implicitly or explicitly what order the claimed components are previously in, whether a claimed component precedes or follows another claimed component, or what order the steps of a manufacturing method are in. The sole purpose of the ordinal numbers is to distinguish a claimed component from another claimed component when the two claimed components are named exactly the same as each other.

(10) Direction-related prepositions and adjectives used herein, such as "on", "upper", "above", "under", "lower" and "below", describe aforesaid two components either in direct contact with each other or not in direct contact with each other.

(11) Features described hereunder in any one of the embodiments of the present disclosure can be

combined to form another embodiment.

Embodiment 1

(12) FIG. 1A is a top view of a portion of a display device according to an embodiment of the present disclosure. FIG. 1B is a cross-sectional view of the display device taken along line M-M' of FIG. 1A. First, this embodiment of the present disclosure provides a substrate **1**. The substrate **1** comprises a display area A and a non-display area B. A photolithography or etching process is performed on the substrate **1** to form a first metal layer (functioning as a first conductive layer **11**) and then on the first conductive layer **11** to form a first insulating layer **12**. The aforesaid steps are repeated to form a second metal layer (functioning as a second conductive layer **13**) and a second insulating layer **14** consecutively. Afterward, a third metal layer (functioning as a third conductive layer **15**) is formed on the second insulating layer **14**, and then a third insulating layer **16** is formed thereon. The third conductive layer **15** comprises a first connection line **151**. The first insulating layer **12** comprises a first hole **121**. The second insulating layer comprises a second hole **141**. The first conductive layer **11** and the first connection line **151** extend into the first hole **121** and the second hole **141**. The first conductive layer **11** and the first connection line **151** contact and electrically connected with each other.

(13) The substrate **1** is a glass substrate, but the present disclosure is not limited thereto. The first conductive layer **11**, the second conductive layer **13**, and the third conductive layer **15** may comprise metal, such as gold, silver, copper, aluminum, molybdenum, titanium, chromium or an alloy thereof, but the present disclosure is not limited thereto. The first connection line **151** may comprises metal or metallic oxide, such as Indium Tin Oxide (ITO) or Indium Zinc Oxide (IZO), or any other conductive material, as needed, but the present disclosure is not limited thereto. The first conductive layer **11**, the second conductive layer **13**, and the third conductive layer **15** may be made of the same material or different materials, provided that the result of a sheet impedance of the first connection line **151** of the third conductive layer **15** divided by a sheet impedance of the first conductive layer **11** electrically connected to the first connection line **151** is greater than 0 and less than or equal to 10. The first conductive layer **11**, the second conductive layer **13**, and the third conductive layer **15** are each independently a monolayer or multilayer structure as needed. The first insulating layer **12**, the second insulating layer **14** and the third insulating layer **16** may comprise silicon oxide, silicon nitride or a combination thereof, but the present disclosure is not limited thereto.

(14) A display device in an embodiment of the present disclosure comprises: a substrate **1** comprising a display area A and a non-display area B; a first conductive layer **11** disposed on the substrate **1** and corresponding to the display area A; a second conductive layer **13** disposed on the substrate **1** and corresponding to the display area A, wherein, the second conductive layer **13** and the first conductive layer **11** cross from top view; a first insulating layer **12** disposed between the first conductive layer **11** and the second conductive layer **13**; a third conductive layer **15** disposed on the substrate **1** and corresponding to the non-display area B, the third conductive layer **15** comprising a first connection line **151**; and a second insulating layer **14** disposed between the second conductive layer **13** and the third conductive layer **15**, wherein, the first conductive layer **11** is electrically connected to the first connection line **151** of the third conductive layer **15**, and the result of a sheet impedance of the first connection line **151** of the third conductive layer **15** divided by a sheet impedance of the first conductive layer **11** electrically connected to the first connection line **151** is greater than 0 and less than or equal to 10. In this embodiment, the non-display area B is surrounded by the display area A. However, in another embodiment, the non-display area B is near the display area A, or the display area A is near just a portion of the non-display area B, as needed. The shape of the non-display area B is not limited by this embodiment and thus may be rectangular, round, polygonal or irregular. The routing of the first conductive layer **11**, second conductive layer **13** or third conductive layer **15** may conform to the shape of the non-display area B.

(15) The first conductive layer **11** functions as a scan line, and the second conductive layer **13**

functions as a data line, but the present disclosure is not limited thereto. The first conductive layer **11** may also function as a data line, or the second conductive layer **13** may also function as a scan line. When the non-display area B comprises a non-circuit area C which are not overlapped with the first conductive layer **11**, the second conductive layer **13** or the third conductive layer **15** but is spaced apart from the display area A by an overly short distance, a short circuit may easily develop between adjacent scan lines or adjacent data lines, because the scan lines or data lines must circumvent the non-circuit area C in order to transmit a signal. On the other hand, the display device cannot have thin bezels, when the distance between the non-circuit area C and the display area A is overly long. According to the present disclosure, the third conductive layer **15** is disposed in the non-display area B, whereas the third conductive layer **15** comprising a first connection line **151** electrically connected to the first conductive layer **11** via the first hole **121** and the second hole **141**; hence, the same signal can be transmitted for a limited distance without causing a short circuit, thereby allowing the display device to have thin bezels. In this embodiment, the third conductive layer **15** is electrically connected to the first conductive layer **11**, so as to transmit signals of the scan lines, but the present disclosure is not limited thereto. In this embodiment, the non-circuit area C can be disposed centrally in the non-display area B and surrounded by the display area A. However, in another embodiment, the non-circuit area C can be near the display area A, or the non-circuit area C is near just a portion of the non-display area B, as needed. The shape of the non-circuit area C is not limited by this embodiment and thus may be rectangular, round, polygonal or irregular. The routing of the first conductive layer **11**, second conductive layer **13** or third conductive layer **15** may conform to the shape of the non-circuit area C.

(16) FIG. 1C is a top view of a portion of the display device according to another embodiment of the present disclosure. The first conductive layer **11** comprises a second wiring **111** and a third wiring **112** electrically insulated with the second wiring **111**. The second wiring **111** is adjacent and parallel to the third wiring **112**. The second wiring **111** and the third wiring **112** extend in a first direction P. The first connection line **151** of the third conductive layer **15** is electrically connected to the second wiring **111**. The projection of the first connection line **151** on the substrate **1** is overlapped with at least a portion of the third wiring **112**.

(17) In another embodiment of the present disclosure, the substrate **1** comprises an opening D disposed corresponding to the non-display area B. The opening D is not overlapped with the first conductive layer **11**, the second conductive layer **13**, or the third conductive layer **15**. The opening D may penetrate the substrate **1**.

(18) This embodiment of the present disclosure is not restrictive of the position of the non-display area B. For instance, the non-display area B can be disposed at the center of the substrate **1**, an edge of the substrate **1**, or a corner of the substrate **1**, but the present disclosure is not limited thereto. In an embodiment of the present disclosure, the display area A of the display device comprises a plurality of pixel units **3**, and the non-display area B is disposed between at least two pixel units **3**. Preferably, each pixel unit **3** is of a width W, and the width of the non-display area B is greater than 2 W.

(19) In another embodiment of the present disclosure, as shown in FIG. 1C, the display area A surrounds the non-display area B, and the first conductive layer **11** comprises a first scan line segment **113**, a second scan line segment **114** and a third scan line segment **115** which extend in the first direction P. The first scan line segment **113** and the second scan line segment **114** are disposed corresponding to two sides of the non-display area B. The third scan line segment **115** is disposed parallel to the first scan line segment **113**. The first connection line **151** of the third conductive layer **15** is electrically connected to the first scan line segment **113** and the second scan line segment **114**. The projection of the first connection line **151** is overlapped with at least a portion of the third scan line segment **115** from top view. The first insulating layer **12** comprises a first through-hole **21** (indicated by a dashed line) and a second through-hole **22** (indicated by a dashed line). The first connection line **151** of the third conductive layer **15** electrically connects to the first

scan line segment **113** via the first through-hole **21**. The first connection line **151** of the third conductive layer **15** electrically connects to the second scan line segment **114** via the second through-hole **22**. In this embodiment, the first connection line **151** and the third scan line segment **115** overlap at two points, namely a junction X, and the first connection line **151** and the third scan line segment **115** are electrically insulated at the junction X. In this embodiment, the third conductive layer **15** and the first conductive layer **11** are not necessarily made of metal. Instead, the first conductive layer **11** may comprise metal, whereas the third conductive layer **15** may comprise metallic oxide, such as ITO or IZO, or any other conductive material or metal, as needed, but the present disclosure is not limited thereto. The display device in this embodiment of the present disclosure will work well and can have thin bezels, provided that limitation on the distances between the wirings in the conductive layers is reduced.

Embodiment 2

(20) FIG. 2A is a top view of a portion of the display device according to another embodiment of the present disclosure. This embodiment is distinguished from embodiment 1 by the technical feature as follows: in this embodiment, the third conductive layer **15** and the second conductive layer **13** are electrically connected, so as to transmit signals of data lines, allowing the display device to have thin bezels. Referring to FIG. 2A, this embodiment has technical features as follows: the third conductive layer **15** comprises a second connection line **152**; the second conductive layer **13** is electrically connected to the second connection line **152** of the third conductive layer **15**; the result of a sheet impedance of the second connection line **152** of the third conductive layer **15** divided by a sheet impedance of the second conductive layer **13** electrically connected to the second connection line **152** is greater than 0 and less than or equal to 10. FIG. 2B is a cross-sectional view of the display device taken along line N-N' of FIG. 2A. As shown in the diagram, the second connection line **152** of the third conductive layer **15** extends into the second hole **141** of the second insulating layer **14** and thus electrically connects to the second conductive layer **13**.

(21) In another embodiment of the present disclosure, the second conductive layer **13** comprises a fourth wiring **131** and a fifth wiring **132** electrically insulated with the fourth wiring **131**. The fourth wiring **131** is adjacent and parallel to the fifth wiring **132**. The second connection line **152** of the third conductive layer **15** is electrically connected to the fourth wiring **131**. The projection of the second connection line **152** on the substrate **1** is overlapped with at least a portion of the fifth wiring **132**.

(22) The materials which the substrate **1**, the first conductive layer **11**, the second conductive layer **13**, the third conductive layer **15**, the first insulating layer **12**, the second insulating layer **14**, and the third insulating layer **16** are made of and the positions of the display area A and non-display area B in this embodiment are the same as their counterparts in Embodiment 1 and thus are, for the sake of brevity, not described herein again. In this embodiment, the first conductive layer **11** functions as a scan line, and the second conductive layer **13** functions as a data line, but the present disclosure is not limited thereto. The first conductive layer **11** can also function as a data line, or the second conductive layer **13** can also function as a scan line.

(23) In another embodiment of the present disclosure, as shown in FIG. 2A, the display area A surrounds the non-display area B, and the second conductive layer **13** comprises a first data line segment **133**, a second data line segment **134** and a third data line segment **135** which extend in a second direction Q. The first data line segment **133** and the second data line segment **134** are disposed corresponding to two sides of the non-display area B. The third data line segment **135** is parallel to the first data line segment **133**. The second connection line **152** of the third conductive layer **15** is electrically connected to the first data line segment **133** and the second data line segment **134**. The projection of the second connection line **152** is overlapped with at least a portion of the third data line segment **135** from top view. The second insulating layer **14** comprises a third through-hole **23** (indicated by a dashed line) and a fourth through-hole **24** (indicated by a dashed

line). The second connection line **152** of the third conductive layer **15** electrically connects to the first data line segment **133** via the third through-hole **23**. The second connection line **152** of the third conductive layer **15** electrically connects to the second data line segment **134** via the fourth through-hole **24**. In this embodiment, the second connection line **152** and the third data line segment **135** overlap at two points, namely a junction Y, and the second connection line **152** and the third data line segment **135** are electrically insulated at the junction Y. In this embodiment, the third conductive layer **15** and the second conductive layer **13** are not necessarily made of metal. Instead, the second conductive layer may comprise metal, whereas the third conductive layer **15** may comprise metallic oxide, such as ITO or IZO, or any other conductive material or metal, as needed, but the present disclosure is not limited thereto. The display device in this embodiment of the present disclosure will work well and can have thin bezels, provided that limitation on the distances between the wirings in the conductive layers is reduced, or the distance between the opening D and the display area A is reduced.

Embodiment 3

(24) FIG. 3 is a cross-sectional view of a portion of the display device according to an embodiment of the present disclosure. This embodiment is distinguished from embodiment 1 by technical features as follows: the first conductive layer **11** and the third conductive layer **15** are electrically connected by the first conductive layer **11** is electrically connected to a portion of the second conductive layer **13**, and the third conductive layer **15** is electrically connected to the portion of the second conductive layer **13** via the second hole **141**, and the second conductive layer **13** is electrically connect to the first conductive layer **11** via the first hole **121**, thus the first conductive layer **11** is electrically connected to the third conductive layer **15**, so as to transmit the same signal, hence, the same signal can be transmitted for a limited distance without causing a short circuit, thereby allowing the display device to have thin bezels.

Embodiment 4

(25) FIG. 4 is a cross-sectional view of a portion of the display device according to another embodiment of the present disclosure. Referring to FIG. 1A and FIG. 4, first, this embodiment of the present disclosure provides a substrate **1**. The substrate **1** comprises a display area A and a non-display area B. A photolithography or etching process is performed to form a third metal layer (functioning as the third conductive layer **41**) in the non-display area B of the substrate **1** and then form a second insulating layer **42** on the third conductive layer **41**. The aforesaid steps are repeated to form a first metal layer (functioning as a first conductive layer **43**), a first insulating layer **44**, a second metal layer (functioning as a second conductive layer **45**), and a third insulating layer **46** consecutively. The first conductive layer **43** is disposed between the first insulating layer **44** and the second insulating layer **42**. The first conductive layer **43** and the third conductive layer **41** are electrically connected through a third hole **5**. In some embodiments (figure not shown), the third conductive layer **41**, the first conductive layer **43** and the second conductive layer **45** are electrically connect to each other. This embodiment is distinguished from Embodiment 1 by the technical feature: this embodiment involves forming a third metal layer first and then forming the other layers consecutively.

(26) In this embodiment, the first conductive layer **43** functions as a scan line, and the second conductive layer **45** functions as a data line, but the present disclosure is not limited thereto. The first conductive layer **43** may also function as a data line, or the second conductive layer **45** may also function as a scan line. In this embodiment, the purpose of the third conductive layer **41** is to reduce limitation on the distances between the wirings in the conductive layers, reduce the distance between the opening D and the display area A, or enable the display device to have thin bezels.

(27) In conclusion, the display device of the present disclosure has thin bezels, by forming a third metal layer in a non-display area, allowing the third metal layer to electrically connect to a first metal layer or a second metal layer, so as to transmit the same signal. The aforesaid embodiments can be combined, but the present disclosure is not restrictive of combinations of the aforesaid

embodiments.

(28) The display device of the present disclosure is applied to various types of display devices comprising an organic light-emitting diode (OLED), a quantum dot (QD), a fluorescence molecule, a phosphor, a light-emitting diode (LED), a micro light-emitting diode (micro LED) or any other display medium. In addition, the display device of the present disclosure is applied to any touch display device equipped with a touch panel. Furthermore, the display device of the present disclosure is applied to any tiled display device, any non-curved, curved or flexible display device or any touch display device.

(29) The aforesaid specific embodiments of the present disclosure must be interpreted to be solely illustrative rather than restrictive, in any way, of the present disclosure.

Claims

1. A display device, comprising: a substrate comprising a display area and a non-display area adjacent to the display area; a first conductive layer disposed on the substrate and comprising a first conductive line; a first insulating layer disposed on the first conductive layer; a second conductive layer disposed on the first insulating layer and comprising a second conductive line, wherein, within the display area, the second conductive line and the first conductive line cross from a top view; a second insulating layer disposed on the second conductive layer; and a third conductive layer disposed on the second insulating layer and comprising a third conductive line; wherein the first conductive line comprises a scan line within the display area, and the second conductive line comprises a data line within the display area, further wherein, within the non-display area, a portion of a projection of the third conductive line on the substrate is overlapped with a portion of a projection of the first conductive layer on the substrate, and within the non-display area, the first conductive layer is electrically connected to the second conductive layer via a hole.
2. The display device of claim 1, wherein, within the non-display area, the third conductive layer is electrically connected to the second conductive layer via another hole.
3. The display device of claim 1, wherein the first conductive line is used as the scan line.
4. The display device of claim 1, wherein the second conductive line is used as the data line.
5. The display device of claim 1, wherein the first conductive layer, the second conductive layer, and the third conductive layer comprise the same materials.
6. The display device of claim 1, wherein the first conductive layer and the second conductive layer comprise different materials, or wherein the first conductive layer and the third conductive layer comprise different materials, or wherein the second conductive layer and the third conductive layer comprise different materials, or wherein the first conductive layer and the second conductive layer and the third conductive layer comprise different materials.
7. The display device of claim 1, wherein the first conductive layer is made of a multilayer material.
8. The display device of claim 1, wherein the second conductive layer is made of a multilayer material.
9. The display device of claim 1, wherein the third conductive layer is made of a multilayer material.
10. The display device of claim 1, wherein a sheet impedance of the third conductive line of the third conductive layer divided by a sheet impedance of the first conductive layer electrically connected to the third conductive line is greater than 0 and less than or equal to 10.
11. The display device of claim 1, wherein a connection between the first conductive layer and the second conductive layer is not in a circuit region.
12. The display device of claim 1, wherein a connection between the second conductive layer and the third conductive layer is not in a circuit region.
13. The display device of claim 1, wherein a connection between the first conductive layer and the

third conductive layer is not in a circuit region.

14. The display device of claim 1, wherein the third conductive layer comprises Indium Tin Oxide (ITO).

15. The display device of claim 1, wherein the second conductive layer comprises a fourth conductive line and a fifth conductive line electrically insulated from the fourth conductive line, further wherein the fourth conductive line is adjacent and parallel to the fifth conductive line, further wherein the fourth conductive line and the fifth conductive line are in the non-display area.

16. The display device of claim 15, wherein the third conductive layer is electrically connected to the fourth conductive line in the non-display area.

17. The display device of claim 15, wherein the projection of the second conductive line on the substrate is overlapped with at least a portion of the fifth conductive line in the non-display area.

18. A display device, comprising: a substrate comprising a display area and a non-display area adjacent to the display area; a first conductive layer disposed on the substrate and comprising a first conductive line; a first insulating layer disposed on the first conductive layer; a second conductive layer disposed on the first insulating layer and comprising a second conductive line, wherein, within the display area, the second conductive line and the first conductive line cross from a top view; a second insulating layer disposed on the second conductive layer; and a third conductive layer disposed on the second insulating layer and comprising a third conductive line; wherein the first conductive line comprises a data line within the display area, and the second conductive line comprises a scan line within the display area, further wherein, within the non-display area, a portion of a projection of the third conductive line on the substrate is overlapped with a portion of a projection of the first conductive layer on the substrate, and within the non-display area, the first conductive layer is electrically connected to the second conductive layer via a hole.

19. The display device of claim 18, wherein, within the non-display area, the third conductive layer is electrically connected to the second conductive layer via another hole.

20. The display device of claim 18, wherein the first conductive layer, the second conductive layer, and the third conductive layer comprise the same materials.

21. The display device of claim 18, wherein the first conductive layer and the second conductive layer comprise different materials, or wherein the first conductive layer and the third conductive layer comprise different materials, or wherein the second conductive layer and the third conductive layer comprise different materials, or wherein the first conductive layer and the second conductive layer and the third conductive layer comprise different materials.

22. The display device of claim 18, wherein the first conductive layer is made of a multilayer material, or wherein the second conductive layer is made of a multilayer material, or wherein the third conductive layer is made of a multilayer material, or wherein the first conductive layer and the second conductive layer are made of multilayer materials, or wherein the first conductive layer and the third conductive layer are made of multilayer materials, or wherein the second conductive layer and the third conductive layer are made of multilayer materials, or wherein the first conductive layer and the second conductive layer and the third conductive layer are made of multilayer materials.

23. The display device of claim 18, wherein a connection between the first conductive layer and the second conductive layer is not in a circuit region.

24. The display device of claim 18, wherein a connection between the first conductive layer and the third conductive layer is not in a circuit region, or wherein a connection between the second conductive layer and the third conductive layer is not in a circuit region, or wherein a connection between the first conductive layer and the third conductive layer is not in a circuit region and a connection between the second conductive layer and the third conductive layer is not in a circuit region.

25. The display device of claim 18, wherein the third conductive layer comprises Indium Tin Oxide (ITO).

26. The display device of claim 18, wherein the second conductive layer comprises a fourth conductive line and a fifth conductive line electrically insulated from the fourth conductive line, further wherein the fourth conductive line is adjacent and parallel to the fifth conductive line, further wherein the first conductive line and the fifth conductive line are in the non-display area.
27. The display device of claim 26, wherein the third conductive layer is electrically connected to the fourth conductive line in the non-display area.
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